Isabella Gilarranz Gimenez

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EDUCATION

University of Miami — Coral Gables, FL *Expected Dec 2027*

B.S. Electrical Engineering & B.S. Computer Science, Minor in Mathematics

Relevant Coursework: Computer Organization & Architecture, Circuit Analysis, Data Structures & Algorithms, Theory of Computation, Intro to AI

Honors: President's Scholarship

Leadership: Administrator, UMaker (UM's student makerspace)

Broward College — Robert "Bob" Elmore Honors College *2023 – 2024*

Associate of Arts, Computer Science

Honors: High Honors, President's List

Leadership: President, STEM Club • VP of Leadership, Phi Theta Kappa Honor Society (Mu Mu Chapter)

EXPERIENCE

UM Robotics and Mechatronics Lab — Undergraduate Researcher *Coral Gables, FL | May 2026 – Present*

- Selected as one of four undergraduate researchers on the lab's founding team; co-lead assembly of the Berkeley Humanoid Lite, an open-source humanoid robot, with a fellow researcher.
- Built and maintain a master tracker consolidating build status and lifecycle data for all 22 actuators, and manage parts procurement, verifying orders against the bill of materials and tracking vendor lead times.
- Contribute to ongoing projects in the lab including a Gemma-based goggle translator, the lab website, and the open-source XLeRobot dual-arm mobile robot.

Safra National Bank of New York — Data Analyst, Business Intelligence *Aventura, FL | Dec 2024 – May 2025*

- Joined the founding team of the bank's newly formed Business Intelligence group; built and maintained 7 Power BI and Tableau dashboards serving 10 departments, writing advanced SQL to extract and structure a decade of banking data (2014–2025).
- Partnered with department management to define KPIs and reporting requirements, translating business questions into dashboard metrics.

Safra National Bank of New York — IT Help Desk Intern *Aventura, FL | May – Dec 2024*

- Diagnosed and resolved hardware and software failures on end-user workstations; coordinated with network admins and hardware vendors.

SBSHS Professional Enhancement Program — Brain-Machine Interface Research Intern *Davie, FL | Feb – May 2024*

- Analyzed EEG signals from behavioral experiments in mice using Python; presented findings at the Annual Pillars Conference.

Broward College — Academic Tutor, Technology Learning Center *Pembroke Pines, FL | Jun 2023 – May 2024*

- Tutored 80+ students in computer science and mathematics courses, including Java and Python programming projects.

PROJECTS

MotionSense_IMU — Custom USB-C IMU Sensor Board | KiCad / USB-C / I²C *2026*

- Designed a USB-C-powered 6-axis IMU board (LSM6DSM) in KiCad: AP2112K 3.3 V LDO power stage, CC pull-down resistors for USB-C sink detection, I²C pull-ups and decoupling network sized from datasheet analysis.
- Routed a 2-layer layout with bottom-layer ground plane pour; verified the design with ERC, DRC, and netlist checks against the schematic.

Portable MP3 Player | ESP32 / C++ / I2S *2025 – Present*

- Designed and built a working handheld MP3 player from scratch: I2S digital audio output to a DAC, TFT display UI, SD card file storage, and tactile controls.
- Wrote all firmware in C/C++ — display driver integration, audio decoding pipeline, file navigation; battery power integration and a custom KiCad PCB revision in progress.

TECHNICAL SKILLS

Hardware & Embedded: KiCad, PCB schematic capture & layout, ESP32 firmware, I²C / SPI / I2S, TFT display drivers, soldering & breadboard prototyping, digital multimeter

Digital Design: Verilog HDL (coursework), digital logic fundamentals, finite state machines

Programming: C/C++, Python (NumPy), SQL, Java, Git

Data & Analytics: Power BI, Tableau, Excel, SQL data extraction & cleaning, KPI dashboard development

Languages: Fluent English & Spanish